Credit Name: CSE 2110 Procedural Programming 1
Assignment Name: CountVowels.java
The following questions can help you in thinking critically about your problem-solving processes:
Understanding the Problem
How did you approach understanding the challenge?
I realized I needed a way to look at every single character inside a string one by one and test whether it matched any of the five standard vowels.
Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?
Checking for both uppercase and lowercase vowels separately seemed repetitive, but I resolved it by converting the entire input string to lowercase right at the start.
Planning the Solution
Did you create a plan or break the problem into smaller steps before coding?
Yes, I broke it into getting the user text, converting it to lowercase, setting up a loop to scan each character, and maintaining a counter variable.
How did you decide on the tools, data structures, or algorithms to use?
I used length() and charAt() string methods[cite: 8], a for loop for iteration[cite: 8], and an if statement with logical ORs
Implementation
Did you write the code in small pieces or attempt the entire solution at once?
I wrote it in pieces: first setting up the Scanner and string input, then writing the for loop structure, and finally adding the vowel validation condition
How did you test your solution along the way to make sure it was working?
I tested it with sentences containing mixed uppercase and lowercase vowels, as well as words with zero vowels, to confirm the counter stayed accurate
Overcoming Challenges
What part of the problem was the most difficult for you?
Making sure the loop conditions didn't cause an out-of-bounds error when checking the last character of the string
How did you handle moments when you felt stuck or unsure of what to do next?
I reviewed how string indexing works in Java (starting at 0 and ending at length() - 1) to ensure my loop index < operator was correct
Learning
Was there anything you learned that you think will help you with future challenges?
I gained a lot of practice looping through strings character by character and using accumulator variables to track counts